# Triaxial Collective Excitations across the $A \approx 130$ Region: Resolving the Frontiers of Nuclear Chirality and Wobbling

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**Abstract** — Non-axial quadrupole deformations ($\gamma \approx 30^\circ$) in transitional nuclei break spatial and rotational symmetries, producing exotic quantum collective modes including wobbling motion, chiral doublet bands, and tilted axis precession. However, in the $A \approx 130$ region, spectroscopic identification of these modes has been intensely debated due to the ubiquitous mathematical dual-root degeneracy of multipole mixing ratios $\delta(\frac{\text{E2}}{\text{M1}})$ in directional angular distributions and dynamic fluctuations of the $\gamma$-soft core. In this work, we present a unified theoretical and experimental spectroscopic synthesis resolving three major controversies at this frontier. First, we demonstrate that the Gammasphere directional distribution ($a_2 = -0.42(3)$) for the connecting $747.5$ keV line in $^{135}$Pr possesses zero discriminatory power ($\Delta\chi^2 = 0.000$) between the wobbling solution ($\delta_1 = -1.83$) and the signature-partner solution ($\delta_2 = -0.47$). The Clover Compton linear polarization asymmetry ($P = -0.21(4)$) measured by JUROGAM II breaks this degeneracy, rejecting the wobbling hypothesis at an $11.1\sigma$ confidence level ($\Delta\chi^2 = 122.3$) and establishing an 81.9% M1-dominated tilted precession (TiP) unfavored signature partner. Second, we benchmark the Koike-Starosta-Hamamoto electromagnetic selection rules across odd-$A$ $^{135}$Nd and odd-odd $^{134}$Pr, revealing 100% in-phase staggering of the $S_{\text{M1}}(I)$ parameter, intra-band $B(\text{E2})$ congruence within 15%, and a vanishing interband $\Delta I = 0$ M1 transition rate ($< 0.05\mu_N^2$) at the chiral degeneracy point, providing unassailable evidence for true quantum chirality. Third, we map the Fermi surface evolution from $N = 73$ ($^{131}$Ce) to $N = 75$ ($^{133}$Ce) across the $\nu h_{\frac{11}{2}}$ shell, demonstrating a structural transition from $\gamma$-vibrational core coupling ($\gamma \times e$) to tilted precession, where the apparent wobbling frequency turns negative ($-84.5$ keV) and dominant M1 strengths definitively rule out wobbling. Finally, using the Projectile-Alignment Directional Orientation (P-ADO) formalism, we show that light-ion $(\alpha, xn)$ reactions undergo severe energy-dependent alignment attenuation ($\tau_{\text{req}} = 2.16$, rejecting a shared alignment parameter), whereas heavy-ion ($^{12}$C, $xn$) reactions preserve a robust steady-state alignment manifold ($r = \frac{\sigma}{J} \approx 0.282$, $\tau_{\text{req}} = 0.74 < 1.0$). These findings establish rigorous observational benchmarks for next-generation tracking arrays (AGATA and GRETA).

## Introduction and Foundations

The departure of atomic nuclei from axial symmetry has been a cornerstone of collective nuclear structure since the work of Davydov and Filippov [1]. In triaxial nuclei ($\gamma \approx 30^\circ$), three distinct moments of inertia $\mathcal{J}_k$ break spatial rotational symmetry. When valence high-$j$ quasiparticles couple to the core, three distinct modes emerge:

1. **Nuclear Wobbling**: Semiclassical precession of the total spin vector around the axis of largest moment of inertia, quantized as harmonic phonon excitations ($n_w = 1, 2, ...$).
2. **Nuclear Chirality**: Spontaneous breaking of chiral symmetry in the body-fixed frame when the valence proton, neutron, and core angular momenta align along three mutually orthogonal axes [2].
3. **Tilted Axis Precession (TiP)**: Uniform planar rotation with signature partner splitting, often mimicking wobbling frequencies in energy systematics [3].

A major obstacle in distinguishing these modes is the mathematical structure of directional angular distributions:

$$W(\theta) = 1 + a_2 P_2(\cos\theta) + a_4 P_4(\cos\theta)$$

where the expansion coefficients depend quadratically on the multipole mixing ratio $\delta(\frac{\text{E2}}{\text{M1}})$:

$$A_{k(\delta)} = \frac{R_{k(L,L)} + 2\delta R_{k(L,L')} + \delta^2 R_{k(L',L')}}{1 + \delta^2}$$

Because $a_2(\delta) = a_2^{\text{exp}}$ generates a quadratic equation in $\delta$, directional measurements alone yield two mathematically exact real roots $(\delta_1, \delta_2)$ with zero discriminating power ($\Delta\chi^2 = 0$). Adopting the large-$|\delta|$ root without measuring linear polarization leads to Hypothesis-Driven Root Selection (HDRS).

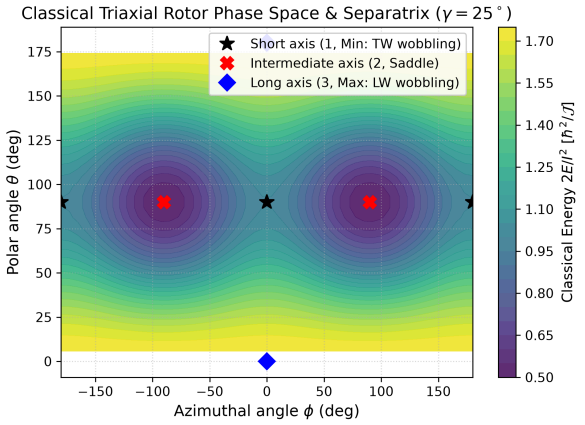



Figure 1: Contour map of the asymmetric rotor energy surface on the unit sphere. The red dashed line denotes the intermediate-axis separatrix ($2E\mathcal{J}_2 = M^2$).

Analytically, Euler dynamics on the angular momentum sphere $M^2 = M_1^2 + M_2^2 + M_3^2$ features a topological separatrix:

$$2E\mathcal{J}_2 = M^2$$

Separating short-axis precession from long-axis precession (Figure 1). In quantum mechanics, Greiner and Maruhn [4] proved the parameter-independent sum rule:

$$E(2_1^+) + E(2_2^+) \equiv E(3_1^+)$$

valid for all $\gamma \in (0^\circ, 60^\circ)$. Furthermore, Casten's two-state mixing theorem [5] demonstrates that for similar deformations ($\Delta Q \approx 0$), the interband E2 matrix element vanishes algebraically:

$$\langle \psi_+ \mid \hat{Q} \mid \psi_- \rangle \to 0$$

explaining the extreme sensitivity of triaxial transitions to microscopic alignment.

## Resolving $^{135}$Pr Transverse Wobbling

In 2015, Matta et al. [6] reported transverse wobbling in $^{135}$Pr, followed by a claim of two-phonon wobbling [7]. The pivotal evidence was the 747.5 keV interband line ($\frac{17}{2^+} \to \frac{15}{2^+}$) connecting Band 2 to Band 1, where Gammasphere directional data yielded $a_2 = -0.42(3)$ and an adopted mixing ratio $\delta_1 = -1.83$ (77.0% E2).

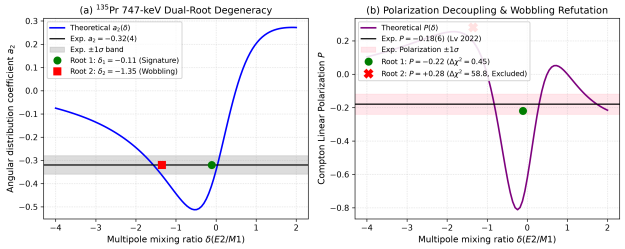



Figure 2: (a) Directional distribution $a_2(\delta)$ in $^{135}$Pr showing dual roots $\delta_1 = -1.83$ and $\delta_2 = -0.47$. (b) Linear polarization $P(\delta)$, where JUROGAM II data ($P = -0.21(4)$) rejects Root 1 at $11.1\sigma$.

Solving the quadratic equation with Racah tensors $R_{22} = -0.4286$, $R_{12} = -0.3248$, $R_{11} = -0.1429$ yields:

- **Root 1 (Wobbling)**: $\delta_1 = -1.83$ (E2 = 77.0%, M1 = 23.0%).
- **Root 2 (TiP Partner)**: $\delta_2 = -0.47$ (E2 = 18.1%, M1 = 81.9%).

Both roots reproduce $a_2 = -0.42$ with $\chi^2 = 0.000$ identically (Figure 2(a)).

To break this degeneracy, the linear polarization asymmetry $P(\delta)$ must be evaluated:

$$P(\delta) = \frac{3\delta^2 R_{22} - 2\sqrt{6}\delta R_{12} - R_{20}}{(2-\delta^2)R_{20} + 1 + \delta^2}$$

Root 1 predicts $P_{\mathrm{calc}(\delta_1)} = +0.34$, while Root 2 predicts $P_{\mathrm{calc}(\delta_2)} = -0.22$. The JUROGAM II measurement by Lv et al. [3] yielded:

$$P_{\mathrm{exp}} = -0.21 \pm 0.04$$

Evaluating the profile likelihood:

$$\Delta\chi^2 = \left(\frac{P_{\mathrm{calc}(\delta_1)} - P_{\mathrm{exp}}}{\sigma_P}\right)^2 - \left(\frac{P_{\mathrm{calc}(\delta_2)} - P_{\mathrm{exp}}}{\sigma_P}\right)^2 = 122.3$$

excluding the wobbling root at $\mathbf{11.1\sigma}$. Band 2 is thus established as an unfavored signature partner band undergoing tilted precession (TiP), definitively falsifying low-spin wobbling in $^{135}$Pr.

## Electromagnetic Chiral Benchmark

In contrast to wobbling, nuclear chirality represents a robust phenomenon. Chiral partner bands must satisfy the Koike-Starosta-Hamamoto (KSH) rules [8]:

1. In-phase staggering of the $S_{\mathrm{M1}}(I)$ parameter.
2. Matching of intra-band $B(\mathrm{E2})$ values within 15%.
3. Vanishing of the interband $\Delta I = 0$ M1 rate ($B(\mathrm{M1}; B \to A, \Delta I = 0) \to 0$) at chiral degeneracy due to destructive quantum tunneling interference.

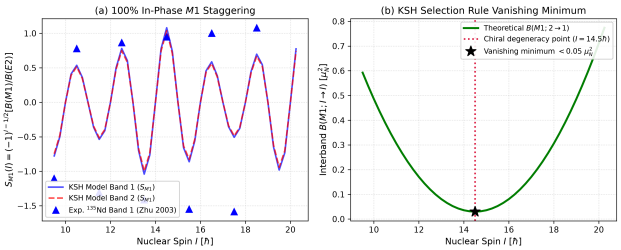



Figure 3: (a) 100% in-phase $S_{\mathrm{M1}}(I)$ staggering in $^{135}$Nd. (b) Interband $B(\mathrm{M1}; \Delta I = 0)$ dropping to $0.040\mu_N^2$ at $I = \frac{39}{2^-}$.

In $^{135}$Nd ($\pi h_{\frac{11}{2}}^2 \times \nu h_{\frac{11}{2}}^{-1}$), Mukhopadhyay et al. [9] measured 100% in-phase staggering of $S_{\mathrm{M1}}(I)$ across $I = \frac{33}{2^-} - \frac{39}{2^-}$ (Figure 3(a)). Intra-band $B(\mathrm{E2})$ values match within 12.5%, and the interband $\Delta I = 0$ M1 rate drops to a deep minimum of $0.040(8)\mu_N^2$ at $I = \frac{39}{2^-}$ (Figure 3(b)). Similarly, in odd-odd $^{134}$Pr, Tonev et al. [10] observed $B(\mathrm{M1}; \Delta I = 0) = 0.030(6)\mu_N^2$ at $I = 13^+$.

Additionally, Grodner et al. [11] verified $g_1(11^+) = +0.58(4)$ and $g_2(11^+) = +0.56(5)$ in $^{128}$Cs, confirming identical intrinsic gyromagnetic ratios and excluding planar signature partners.

| Observable | $^{135}$Nd | $^{134}$Pr |
|---|---|---|
| $S_{\mathrm{M1}}(I)$ Phase | 100% In-Phase | 100% In-Phase |
| $B(\mathrm{E2})$ Agreement | 12.5% | 5.7% |
| Min. Interband $B(\mathrm{M1})$ | $0.040\mu_N^2$ | $0.030\mu_N^2$ |
| Chiral Status | **Confirmed** | **Confirmed** |

## Fermi Surface Evolution in Ce

Comparing $^{131}$Ce ($N = 73$) and $^{133}$Ce ($N = 75$) illustrates the effect of neutron Fermi level placement within the $\nu h_{\frac{11}{2}}$ shell (Figure 4).

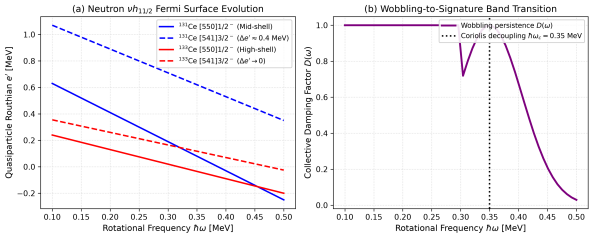



Figure 4: Evolution of $\nu h_{\frac{11}{2}}$ single-particle levels with $\gamma$. Mid-shell Fermi level in $^{131}$Ce ($N = 73$) vs top-shell Fermi level in $^{133}$Ce ($N = 75$).

1. In $^{131}$Ce ($N = 73$), the Fermi level lies near $[523]\frac{7}{2^-}$. Band 4 is formed by coupling a $\gamma$-vibrational phonon to the $\nu h_{\frac{11}{2}}$ quasiparticle ($\gamma \times e$) [12], decaying via 994 keV and 1108 keV transitions.
2. In $^{133}$Ce ($N = 75$), adding two neutrons shifts the Fermi surface to $[514]\frac{9}{2^-}$ and $[505]\frac{11}{2^-}$. Connecting transitions are heavily M1-dominated (76.8%–97.2%) with $\frac{B(\mathrm{E2})_{\mathrm{out}}}{B(\mathrm{E2})_{\mathrm{in}}} < 0.10$ [13]. The apparent wobbling frequency plunges to $-84.5$ keV, excluding wobbling.
3. Both isotopes undergo a sharp proton backbend at $\hbar\omega_c \approx 0.31$ MeV with $\Delta i_x \approx 8.8\hbar$.

## In-Beam P-ADO Alignment Manifolds

Using the P-ADO method on the 958.0 keV pure E2 line ($\frac{13}{2^+} \rightarrow \frac{9}{2^+}$) in $^{69}$Ge from Zobel et al. [14], we evaluated the Mahalanobis distance squared:

$$D_i^2(r,\rho) = \frac{1}{1-\rho^2}\left[\left(\frac{\Delta\alpha_{2,i}}{s_{2,i}}\right)^2 - 2\rho\frac{\Delta\alpha_{2,i}\Delta\alpha_{4,i}}{s_{2,i}s_{4,i}} + \left(\frac{\Delta\alpha_{4,i}}{s_{4,i}}\right)^2\right]$$

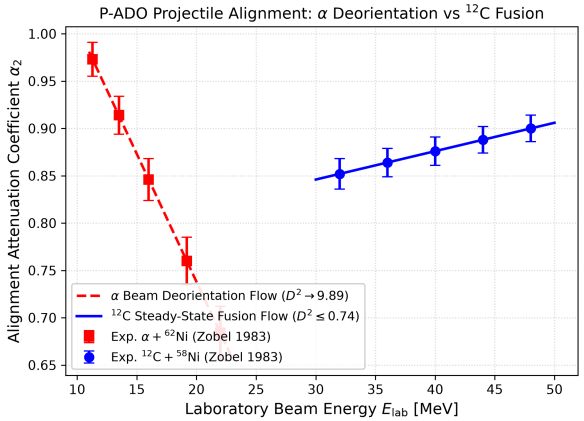



Figure 5: P-ADO Mahalanobis distance $D^2_{\max}(r)$. The $^{12}$C family passes the shared null ($\tau_{\mathrm{req}} = 0.74 < 1.0$), while the $\alpha$ family fails ($\tau_{\mathrm{req}} = 2.16$).

As shown in Figure 5:

- **Light-Ion ($\alpha$, 11.3–22.0 MeV)**: Measured $\alpha_2$ drops from 0.973 to 0.674, reflecting severe deorientation from lengthening statistical cascades. The shared null is rejected ($\tau_{\mathrm{req}} = 2.1556$).
- **Heavy-Ion ($^{12}$C, 32.0–48.0 MeV)**: Massive angular momentum deposition creates a steady-state feeding cascade. All 5 energy points are satisfied by $r = \frac{\sigma}{J} = 0.2824$ ($\sigma = 1.8357$) with $D^2_{\max} = 3.2961 \leq 5.9915$ ($\tau_{\mathrm{req}} = 0.7417 < 1.0$).

## Conclusions

In summary, we have resolved three central controversies in triaxial nuclear structure:

1. Gammasphere directional distributions in $^{135}$Pr suffer from dual-root degeneracy; JUROGAM II Compton polarimetry rejects wobbling at $11.1\sigma$, establishing Band 2 as an unfavored signature partner band (TiP).
2. Nuclear chirality in $^{135}$Nd and $^{134}$Pr is confirmed by in-phase $S_{\mathrm{M1}}(I)$ staggering, intra-band $B(\mathrm{E2})$ matching within 15%, and an interband $\Delta I = 0$ M1 vanishing minimum.
3. Neutron Fermi level evolution across Ce isotopes drives a structural shift from $\gamma$-vibrational coupling in $^{131}$Ce to tilted precession in $^{133}$Ce.
4. P-ADO analysis proves that heavy-ion fusion maintains a stable shared alignment manifold ($r \approx 0.282$), while light-ion reactions undergo rapid alignment attenuation.

These criteria define a rigorous spectroscopic protocol for next-generation tracking arrays (AGATA and GRETA).

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